History — Biology Chapter 7: Metabolism

The humans behind the science. Every idea in this chapter came from a real person who stayed up late, argued with colleagues, made lucky mistakes, and sometimes risked everything for an idea. Here are their stories.

Eduard Buchner (1860–1917) — The Man Who Proved Life Wasn't Magic

For most of the 19th century, the great debate in biology was this: can chemistry happen without life? The famous chemist Louis Pasteur insisted that fermentation — the process by which yeast converts sugar to alcohol — could only happen in living cells. It wasn't chemistry; it was "vital force," something special about being alive.

Then Eduard Buchner came along.

Buchner was a German chemist who, in 1897, was trying to make a medical preparation from yeast. He ground up yeast cells, squeezed out the juice, and mixed it with sugar as a preservative. To his astonishment, the yeast juice started bubbling — producing CO_2, just like living yeast would. And there wasn't a living yeast cell in the mixture. He'd killed them all.

He'd just proven that fermentation is plain chemistry. The yeast juice contained **enzymes** — proteins that could carry out the reactions without any living cell required. Life's chemistry is just chemistry.

Buchner called the active substance "zymase" (from the Greek *zyme* = leaven/yeast). He spent years convincing a skeptical scientific community. In 1907, he won the Nobel Prize in Chemistry.

The tragic coda: Buchner was 57 years old in 1917, during World War I. Despite his Nobel Prize, he volunteered for active duty as a supply officer. He was killed by a grenade fragment at the Battle of Romania. One of the founders of biochemistry, dead in a trench.

Emil Fischer (1852–1919) — The Lock-and-Key Man

Emil Fischer was the greatest organic chemist of his era — a man who synthesized sugars, amino acids, and purines with such precision that other chemists compared him to an artist. He also had the misfortune of being surrounded by tragedy.

In the 1890s, Fischer was studying how enzymes interact with their substrates. He noticed that enzymes were remarkably specific — an enzyme that broke down one sugar wouldn't touch a structurally similar sugar. He described this with an elegant metaphor in 1894: the enzyme was a "lock" and the substrate was a "key." Only the right key could fit.

The lock-and-key model became one of the most famous images in biochemistry. It explained enzyme specificity in terms everyone could understand.

Fischer's personal life was far darker than his science. His wife died young. Two of his three sons died in World War I. A third son died of illness shortly after. Fischer himself suffered from cancer and died in 1919, shortly after the war ended, by suicide.

The man who gave us the image of molecular specificity died having lost almost everything. His lock-and-key model outlived all of it.

Hans Krebs (1900–1981) — The Cycle That Bears His Name

Hans Krebs was a German Jewish biochemist who discovered the citric acid cycle — what we now call the **Krebs cycle** — one of the central metabolic pathways in all aerobic life.

In 1933, when the Nazis came to power, Krebs was working at the University of Freiburg. He received a letter dismissing him from his position under the new racial laws. He was Jewish. He had one day to clear out his office.

He fled to England, eventually settling at Cambridge. Far from being crushed, he was extraordinarily productive in exile. In 1937, working in Sheffield, he described the complete cycle of reactions by which pyruvate is oxidized to CO_2 and water in mitochondria — extracting the energy from the carbon bonds and loading it onto electron carriers.

He submitted his findings to the journal *Nature*. They rejected it — too long a wait for publication, they said. He published it in a different journal instead.

In 1953, he won the Nobel Prize in Physiology or Medicine. He remained modest and hard-working until the end. A man driven out of his country by bigotry came back, in a sense, by discovering how every living cell on Earth gets its energy.

Melvin Calvin (1911–1997) — The Carbon Detective

Melvin Calvin grew up in modest circumstances in Minnesota, the son of Russian-Jewish immigrant parents. He went on to become one of the most methodical scientists of the 20th century — and the man who cracked the puzzle of how plants fix carbon dioxide.

In the late 1940s and 1950s, Calvin and his team at Berkeley used a brilliant trick: they fed algae radioactive carbon dioxide ($^{14}CO_2$), then stopped the photosynthesis at precise intervals by killing the cells. They then separated all the molecules in the cells by chromatography and traced which molecules had picked up radioactive carbon at each time point.

By doing this at intervals of seconds, they pieced together the entire sequence of reactions — step by step — that converts CO_2 into glucose. The cycle they discovered is now called the **Calvin cycle**.

Calvin was famously intense. He reportedly could work 18-hour days and expected the same from his students. But he was also genuinely collaborative and gave enormous credit to his team members.

He won the Nobel Prize in Chemistry in 1961 — uniquely, he was the sole recipient, unusually uncommon for a discovery of that scale.

Calvin lived until 1997. In his later years, he worked on using plants for biofuels — wanting plants to produce liquid fuels from sunlight, decades before the concept became fashionable.

Otto Warburg (1883–1970) — The Oxygen Obsessive

Otto Warburg was one of the most dominant and difficult personalities in 20th-century biology. He won the Nobel Prize in Physiology or Medicine in 1931 for discovering that cancer cells respire differently from normal cells — they prefer anaerobic glycolysis even when oxygen is available (now called the "Warburg effect"). He was nominated for a second Nobel and should probably have won a third.

He was also an extremely difficult person to work with — imperious, dismissive, and unwilling to credit others. But his discoveries were real and profound.

The remarkable biographical footnote: Warburg was half-Jewish by Nazi racial laws. He was also a close friend of the Nazi biochemist who persuaded Hitler that Warburg was too valuable to persecute — because Hitler was terrified of cancer, and Warburg was the world's leading cancer researcher. Warburg was thus protected while his relatives were not.

He used this protection to continue working in Berlin throughout the entire war, making important discoveries about photosynthesis — how many photons of light are required per oxygen molecule released — while millions of others with similar heritage were being murdered.

He died in 1970, having refused to retire, still running his institute at 87.

Peter Mitchell (1920–1992) — The Lonely Heretic

The story of how ATP is actually synthesized by the electron transport chain is one of the most dramatic in science.

In 1961, British biochemist Peter Mitchell proposed a radical idea called the **chemiosmotic theory**: that the ETC works not by direct chemistry, but by pumping protons (H^+ ions) across the inner mitochondrial membrane, creating a gradient. Protons then flow back down this gradient through an enzyme (ATP synthase), and the flow of protons drives the synthesis of ATP — like water flowing downhill through a turbine.

The biochemistry establishment rejected this theory completely. It sounded wrong. It didn't fit the prevailing models. Mitchell was seen as a crank.

Mitchell set up his own private research institute in a renovated manor house in Cornwall — Glynn Research Ltd — and continued working, largely ignored. His experiments kept confirming his theory. Slowly, over a decade, the evidence became undeniable. By the 1970s, the scientific consensus had swung to Mitchell's side.

In 1978, he won the Nobel Prize in Chemistry — alone, for a theory he had developed almost entirely outside mainstream institutions, against sustained opposition from the most prestigious names in biochemistry.

The proton gradient he described is the mechanism by which all aerobic life — every animal, every aerobic bacterium, every cell with a mitochondrion — makes its ATP. The loneliest heresy in biochemistry turned out to be the truth.

James Sumner (1887–1955) — The One-Armed Crystallizer

James Sumner grew up wanting to be a chemist. At age 17, while hunting, he had a shooting accident that required the amputation of his left arm. He was right-handed, but his professors at Harvard told him a one-armed person couldn't succeed in laboratory chemistry.

He ignored them and earned his PhD.

In the 1920s, Sumner was working on urease — an enzyme that breaks down urea. He spent nine years attempting to crystallize it (crystallization was the method used to purify and identify proteins). His colleagues thought he was wasting his time — most biochemists didn't even believe enzymes were proteins.

In 1926, he succeeded. He crystallized urease and showed beyond doubt that it was a protein. He was instantly dismissed by the establishment. The renowned biochemist Richard Willstätter declared Sumner's methods too crude to be trusted.

It took until 1946 for Sumner to receive the Nobel Prize in Chemistry — shared with John Northrop who crystallized pepsin and trypsin. By then, the protein nature of enzymes was beyond dispute.

Sumner's response to the years of dismissal was typically dry: "I have always believed that you should not trust anyone who has not actually been wrong at least once."

A Note on the Nameless Thousands

Almost every concept in this chapter — the Krebs cycle, the Calvin cycle, the ETC, ATP synthesis — required not just one genius but dozens of scientists working across decades, sharing data, arguing, replicating, and slowly building consensus.

The names we remember are the peaks. But the mountain they stood on was built by thousands of researchers, many of them never credited, many of them women and non-Western scientists who made essential contributions to biochemistry at a time when recognition was far from equally distributed.

The science here is old enough to celebrate and recent enough to still be unfolding. The story of metabolism is still being written.